

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JULY / AUGUST 2025

Program	: B.E. - Common to CSE / ISE / AI & ML / AI & DS / CSE (AI & ML) / CSE (CY)	Semester	: IV
Course Name	: Statistics, Probability and Linear Programming	Max. Marks	: 100
Course Code	: 23 CS / IS / AI / AD/ CI / CY 41 22 CS / IS / CI / CY41	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Use of Statistical Table is permitted.

UNIT - I

- Define Continuous random variable and give an example. CO1 (02)
 - Out of 520 families with 5 children each, how many would you expect to have (i) 3 boys (ii) 3 girls. Assume the probability of a girl child is 0.45. CO1 (04)
 - A random variable X has the following probability function: CO1 (07)

X	1	2	3	4	5	6	7
$P(X)$	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Find (i) k (ii) $P(X \geq 6)$ (iii) $E(X)$.

- Fit a least square regression plane $Y = a + bX_1 + cX_2$ for the following data: CO1 (07)

Y	9	7	5	4
X_1	30	52	64	82
X_2	16	10	7	4

- Write the equations of two regression lines. CO1 (02)
 - A continuous random variable X has pdf given by $f(x) = \begin{cases} 2e^{-2x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$ CO1 (04)

Evaluate (i) $E(X)$ (ii) $E(X^2)$.

- Dynamic viscosity of water $\mu (10^{-3} \text{ N.s/m}^2)$ is related to temperature $T (^{\circ}\text{C})$ in the following manner. Estimate viscosity of water at $T = 35^{\circ}\text{C}$ by framing a parabolic relation $\mu = aT^2 + bT + c$. CO1 (07)

T	0	5	10	20	30	40
μ	1.787	1.519	1.307	1.002	0.7975	0.6529

- A communication channel receives independent pulses at the rate of 12 pulses per micro second. The probability of transmission error is 0.001 for each micro second. Compute the probability of (i) no error during a micro second (ii) one error per micro second (iii) at least three error per micro second (iv) At most two errors. CO1 (07)

UNIT - II

- Write pdf of Gamma distribution. CO2 (02)
 - On a certain city transport route, buses ply every 30 minutes between 6 a.m. and 10 p.m. If a person reaches a bus stop on this route at a random time during this period, what is the probability that he will have to wait for at least twenty minutes? CO2 (04)

- c) The joint distribution of two variables X and Y are given below CO2 (07)

$X \backslash Y$	-4	2	7
1	1/8	1/4	1/8
5	1/4	1/8	1/8

Find (i) Marginal distributions of X and Y (ii) $COV(X, Y)$

- d) A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live (i) more than 32 months (ii) less than 28 months (iii) between 37 and 49 months. CO2 (07)
4. a) Write mean and variance of uniform distribution. CO2 (02)
 b) Steel rods are manufactured to be 3 cm in diameter but they are acceptable if they are inside the limits 2.99 cm and 3.01 cm. It is observed that 5% are rejected as oversized and 5% are rejected as undersized. Assuming that the diameters are normally distributed, find the standard deviation of the distribution. CO2 (04)
 c) The mileage which car owner get with a certain kind of radial tyre is a random variable having an exponential distribution with mean 30,000km. Find the probabilities that one of these tyres will last (i) at least 15,000km (ii) at most 25,000km. CO2 (07)
 d) A privately owned business operates both drive in facility and a walk in facility. On a randomly selected day let X and Y respectively be the proportions of the time that drive in and walk in facilities are in use and suppose that the joint density function of these random variables is CO2 (07)

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Show that (i) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$ (ii) Find $P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}\right)$.

UNIT - III

5. a) Write the expression for i) expected number of customers in the queue ii) expected number of customers in the systems for $(M/M/s):(\infty/\text{FIFO})$ queuing system. CO3 (02)
 b) Suppose that customers arrive at a Poisson rate of one per every 12 minutes and that the service time is exponential at a rate of one service per 8 minutes. What is the average number of customers in the system. CO3 (04)
 c) There are 2 white marbles in bag A and 3 red marbles in bag B. At each step of the process a marble is selected at random from each bag and the two marbles selected are interchanged then find (i) transition probability matrix (ii) what is the probability that there are 2 red marbles in A after 3 steps (iii) In the long run what is the probability that there are 2 red marbles in A? CO3 (07)
 d) Trains arrive at the yard every 15 minutes and the service time is 33 minutes. If the line capacity of the yard is limited to 4 trains, then find CO3 (07)
 (i) The probability that the yard is empty.
 (ii) The average number of trains in the system.

6. a) Write the expression to determine the expected number of customers in the queue for (M/M/1): (k/FIFO). CO3 (02)
b) Check whether the following Stochastic matrix is Regular or not? CO3 (04)

$$\begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$$

- c) Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate of 20 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer in the queue and in the system. CO3 (07)
d) A petrol pump has 4 pumps. The service time follows an exponential distribution with a mean of 6 minutes and cars arrive for service in a Poisson process at the rate of 30 cars per hour. Find the average waiting time in the queue, average time spent in the system and the average number of cars in the system. CO3 (07)

UNIT- IV

7. a) Define Null Hypothesis and Alternative Hypothesis. CO4 (02)
b) The nine items of a sample have the following values:45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these differ significantly from the assumed population mean of 47.5? Test at 5% LOS. CO4 (04)
c) A sample of size 13 gave an estimated population variance of 3.0, while another sample of size 15 gave an estimate of 2.5. Could both samples be from populations with the same variance? Test at 5% LOS. CO4 (07)
d) The following table shows the result of an experiment to investigate the effect of vaccination induced on the animals against a particular disease. Use χ^2 - test to test the hypothesis that there is no difference between and vaccinated and unvaccinated groups i.e. vaccination and this disease are independent. Test at 5% LOS. CO4 (07)

	Got disease	Did not get disease
vaccinated	9	42
Not vaccinated	17	28

8. a) Define Type-I and Type – II errors. CO4 (02)
b) A sample of 900 members is found to have a mean of 3.4 cm. Can it be reasonably regarded as a truly random sample from a large population with mean 3.25 cm and S.D 1.61 cm. Test at 1% LOS. CO4 (04)
c) Two independent samples of sizes 8 and 7 contained the following values: CO4 (07)

Sample 1	19	17	15	21	16	18	16	14
Sample 2	15	14	15	19	15	18	16	

- Is the difference between the sample means significant? Test at 5%LOS
d) The theory predicts that the proportion of beans available in four groups A, B, C, D should be 9:3:3:1. In an experiment with 1600 beans the numbers in the four groups were 882, 313, 287 and 118. Use χ^2 - test to verify whether the experimental result supports the theory. Test at 5% LOS. CO4 (07)

UNIT - V

9. a) Use Big-M method to maximize $z = 3x_1 + 2x_2$, subject to the constraints, $2x_1 + x_2 \leq 2$, $3x_1 + 4x_2 \geq 12$, $x_1 \geq 0$, $x_2 \geq 0$. CO5 (10)
- b) Solve the following LPP by Simplex method : CO5 (10)
- $Max\ Z = 6x_1 + 11x_2$
 $s/t\ 2x_1 + x_2 \leq 104$
 $x_1 + 2x_2 \leq 76$
 $x_1, x_2 \geq 0$
10. a) Convert the following problem in dual form: CO5 (04)
- Maximize: $z = x + y + z$ subjected to $x - 3y + 4z = 5$, $x - 2y \leq 3$,
 $2y - z \geq 4$, where $x, y \geq 0, z$ unrestricted.
- b) Solve $Min\ z = 5x + 4y$ subject to the constraints $x + 2y \leq 10$, $x + y \geq 8$, $2x + y \geq 12$, $x \geq 0$, $y \geq 0$ by graphical method. CO5 (06)
- c) Solve the following LPP by Two-phase method CO5 (10)
- $Min\ Z = 2x_1 + 3x_2$
 $s/t\ x_1 + x_2 \leq 5$
 $x_1 + 2x_2 \geq 6$
 $x_1, x_2 \geq 0$
